

Profile

A mathematically highly skilled R&D engineer with a pioneering publication record in stochastic geometry modelling of satellite communication systems, including a first-author publication in the high-impact journal *Institute of Electrical and Electronics Engineers Transactions on Communications* (IEEE TCOM) and a first-author presentation at the flagship conference IEEE GLOBECOM. A creative mindset that seeks simplicity – not complexity – and is not afraid to fiddle with the details. Committed to bringing my unique expertise to a collaborative, development-driven, and fast-paced environment, relishing complex challenges at the forefront of the aerospace industry. Always striving to grow professionally and spiritually, and to cultivate compassion and humility in all aspects of life. Ready to relocate; FI/EU citizen.

Skills

• Data analysis and visualisation • Inter-institute collaboration • Project management • Research • Stakeholder management
• 3GPP NR NTN technical standards • High-frequency link budget analysis and performance modelling • RF interference spectrum analysis • Statistical digital signal processing • Trade-off analysis and system optimisation

Experience and Impact

- 5/2025– **Senior Researcher**, *Aalto University*, School of Electrical Engineering, Espoo, Finland
- Managed 2 R&D projects as the corresponding author and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication.
 - Mentored junior researchers as a bachelor's degree supervisor and helped to establish future talents.
 - Voluntary work as a peer reviewer in various journals, including IEEE TCOM, IEEE TWC, IEEE TMC, etc.
 - Put forth a novel closed-form Gaussian autocorrelation function for aggregate random radio frequency (RF) interference at a low Earth orbit (LEO) satellite terminal, applicable to ground-space interference spectrum analysis.
 - Proposed, analysed, and simulated successive interference cancellation (SIC) algorithms for multibeam LEO uplink, improving coverage performance by 900% for the third-strongest Earth transmitter in terms of signal-to-interference ratio (SIR). The SIC improves user fairness and enables serving multiple terrestrial terminals per cell, while preserving the average total network throughput at the LEO satellites, which is optimised around 1 (bit/s/Hz).
- 8/2023– **Research Internship**, *University of Notre Dame*, Dept. of Electrical Engineering, South Bend, IN, US
- 2/2024
- Based on link budget and RF interference analysis, a novel statistical SIR model was proposed by characterising the SIR with a Lomax distribution in a narrow-band, high-frequency LEO uplink. Lomax distribution is a well-known heavy-tailed distribution. Consequently, many well-known results can be utilised for efficient performance analysis and optimisation of LEO communications, and for comprehensive insight.
 - Coordinated a distributed team in a cross-functional project and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication in the high-impact journal IEEE TCOM.
 - Identified an optimal cell size in a LEO uplink that maximises average throughput and quantified a trade-off between average throughput and performance consistency: –10 dB cell size provides high average throughput but highly varying link quality of service (QoS) over the beams, whereas –3 dB cells provide more stable QoS but are detrimental to the throughput.
 - International relations.
 - Secured project financing of 4349 € for personal internship expenses from the Väisälä Foundation.
- 09/2020– **PhD Researcher**, *Aalto University*, Signal Processing of Wireless Networks, Espoo, Finland
- 05/2025
- Built, maintained, and documented a multi-language simulation stack (GIT, MATLAB, MATHEMATICA, OCTAVE, PYTHON) including algorithms for analysis, simulation, and visualisation, used across several research projects.
 - Developed a novel and well-cited stochastic geometry framework for the analysis of LEO networks, reducing simulation time by an order of magnitude 100 times more computationally efficient than the MATLAB Satellite Communication Toolbox while providing equivalent results on the network performance metrics. The project was a collaboration with Technical Research Centre of Finland (VTT).
 - An industry presentation at IEEE GLOBECOM. Based on the presented work, the LEO uplink RF interference can be characterised by a gamma distribution. Consequently, many well-known results for the gamma distribution can be utilised for efficient performance analysis and optimisation, and for comprehensive insight.
 - Proposed, analysed, and simulated a minimal-overhead slotted ALOHA-like inter-satellite link (ISL) data routing algorithm in a constellation of 10 planes, 6 satellites in each plane at the orbital altitude of 1000 km, using graph algorithms. The QoS was optimised at an average latency of 30 ms and total network spectral efficiency 1 (bit/s/Hz).
 - Proposed, analysed, and simulated a minimal-overhead, grant-free, random medium access control (MAC) for multibeam LEO uplink that mitigates the high variation in individual link performance while preserving average network throughput, optimised around 1 (bit/s/Hz).

Education

Expected 2026 **Doctor of Science, PhD**, *Aalto University, School of Electrical Engineering*, Signal processing for communications, • Stochastic geometry analysis and simulation of LEO networks
MSc, *Univ. of Helsinki*, Applied analysis, • Finite element method (FEM) • Helmholtz equation
BSc, *Univ. of Helsinki*, Applied analysis, • Optimal control theory • System performance optimisation

References

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- MSc. Abid Afridi, Aalto University, abid.afridi@aalto.fi